

Recognizing target objects from reach to grasp movements

A plain language report of the RTG experiments

About 15–20 minutes including the figures. This report explains how the Continuous Thought Machine (CTM) was connected to real reach-to-grasp recordings, what it learned, how early it could recognize the target, and what changed when one model was trained for each participant.

The main result is 85.31% recognition accuracy when the model is trained on B0 and tested on B1–B4. Recognition improves as more of the movement becomes available. Using shorter raw inputs gives almost the same results as stretching each available prefix to 100 positions. Separate models help two participants but do not improve the overall result.

Main evaluation	Correct recordings	Accuracy
Pooled CTM most-certain tick	546 / 640	85.31%
Pooled CTM final tick	548 / 640	85.63%
Five participant-specific CTMs most-certain tick	537 / 640	83.91%

The central question is whether movement patterns learned in one block still identify objects in later blocks with changed arrangements. The same five participants occur in training and testing. This is block transfer; it is not a test on completely new people.

How to read the results

The final experiments come first. Earlier experiments appear near the end because they helped develop the pipeline but used a different split. Their 96.25% result is useful context, not the headline score for the final task.

Unless stated otherwise, accuracy uses the CTM prediction at its most-certain internal tick. Percentage points describe the difference between two percentages. All results are from the saved runs; no additional models were trained for this report.

The path through the report

Prepare and label movements → train on B0 → evaluate transfer by block and object → shorten the available movement → check physical time and input length → compare pooled and personal models → connect the saved models to internal-activity analysis.

From recordings to model input

The retained dataset contains 958 successful recordings from five participants and eight object classes. A recording is one movement attempt. Each model input keeps the x, y and z coordinates of five fingertips: 15 channels in total. The sensor and label information used to define that input has a different role from the coordinates given to the classifier.

Data partition	Recordings	Role in the final experiment
B0	318	Training and fitting feature scaling
B1-B4	640	Transfer evaluation
Each test participant	128	Separate participant reporting
Each test object	80	Balanced eight-class evaluation

Object identity and board location are deliberately separated. In B0, the trial number identifies the object and board sensor 3 supplies the lift signal. In B1-B4, the corrected target field identifies the object while the trial number identifies the board sensor. Confusing these two roles would change either the labels or the movement endpoint.

For B1-B4 repetition 0, the known target-field error is corrected using repetition 1 from the same participant, block and trial. This is a metadata correction, not a shift of the movement coordinates and not a blanket addition or subtraction of one. Failed attempts are excluded.

The final transfer representation uses the same crop rule across blocks: start at the first active-hand sample and end at the detected board-displacement crossing. The board signal is zeroed at movement onset; the crossing uses an absolute change exceeding 10% of its maximum subsequent relative displacement. The corresponding crossing sample is retained, with no added post-lift tail.

For training, each cropped trajectory is linearly interpolated over normalized sample index to 100 positions. Each coordinate is then standardized using only B0 means and standard deviations. Test data use those same values. This prevents test-set statistics from silently changing the input representation.

The lift reference is determined retrospectively from the recorded board signal. That is appropriate for evaluating prefixes relative to a known event, but it does not demonstrate an online lift detector. The model itself receives fingertip coordinates, not the board signal or object label.

Training and the meaning of a CTM decision

The final pooled model uses all 318 B0 recordings. No B0 validation subset is held back. Instead, the training duration is fixed at 15 epochs, giving 150 optimizer updates with batches of 32. An epoch means one pass through the training set; an update means one weight adjustment from a batch.

The duration comes from an earlier B0 development split: its selected epoch 19 corresponded to about 152 updates. Preserving approximately that budget lets the final model use all B0 without selecting its training endpoint from the final test score. Earlier project experiments had already examined B1–B4, so the overall project is exploratory rather than a completely untouched confirmatory test.

Model setting	Value
Fingertip input	15 channels × 100 positions during training
Object classes	8
Internal state units	256
Internal computation	25 ticks
State history length	15
Trainable parameters	328,970
Learning rate and seed	0.0003 and seed 0

There are two different clocks. Movement positions describe the external recording. Internal ticks describe repeated computation after the available input is supplied. Twenty-five ticks do not mean 25 new movement samples or a measured number of milliseconds.

At each tick, the CTM produces eight class scores. Its native certainty measures how concentrated the class distribution is, using normalized entropy. The most-certain readout chooses the prediction at the tick with highest certainty; the final-tick readout uses tick 25. Neither requires knowing the correct object to choose that tick.

Certainty is not automatically the probability that the prediction is correct. Also, choosing the most-certain tick after running the full computation is not an early stopping policy. The small difference between the two readouts should not be presented as evidence of reliable adaptive stopping.

The trained model reaches 99.06% on B0. That confirms that it fits its training recordings well. It is not an independent B0 test score and does not by itself measure generalization.

Where transfer succeeds and fails

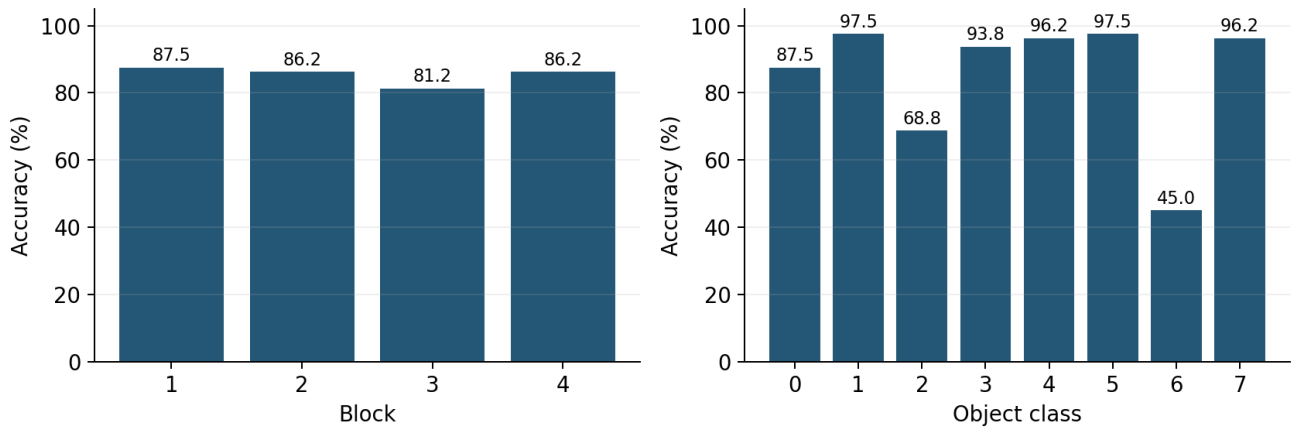


Figure 1. Final pooled-model accuracy by block and object. Each block contains 160 recordings; each object contains 80. All use the same frozen B0-trained model.

Object	Correct / 80	Recall
0	70	87.50%
1	78	97.50%
2	55	68.75%
3	75	93.75%
4	77	96.25%
5	78	97.50%
6	36	45.00%
7	77	96.25%

Performance is fairly similar across blocks: B1 is 87.50%, B2 and B4 are 86.25%, and B3 is 81.25%. The stronger contrast is between objects. Classes 2 and 6 account for 69 of the 94 errors, or about 73%. Class 6 remains difficult in every test block, rather than failing only in B3.

The broad conclusion is successful but uneven transfer. Several objects are recognized very reliably, while a small part of the label space causes most mistakes. This identifies where to look next, but it does not establish whether the cause is movement similarity, label quality, the input representation, or the model.

Reading the errors more closely

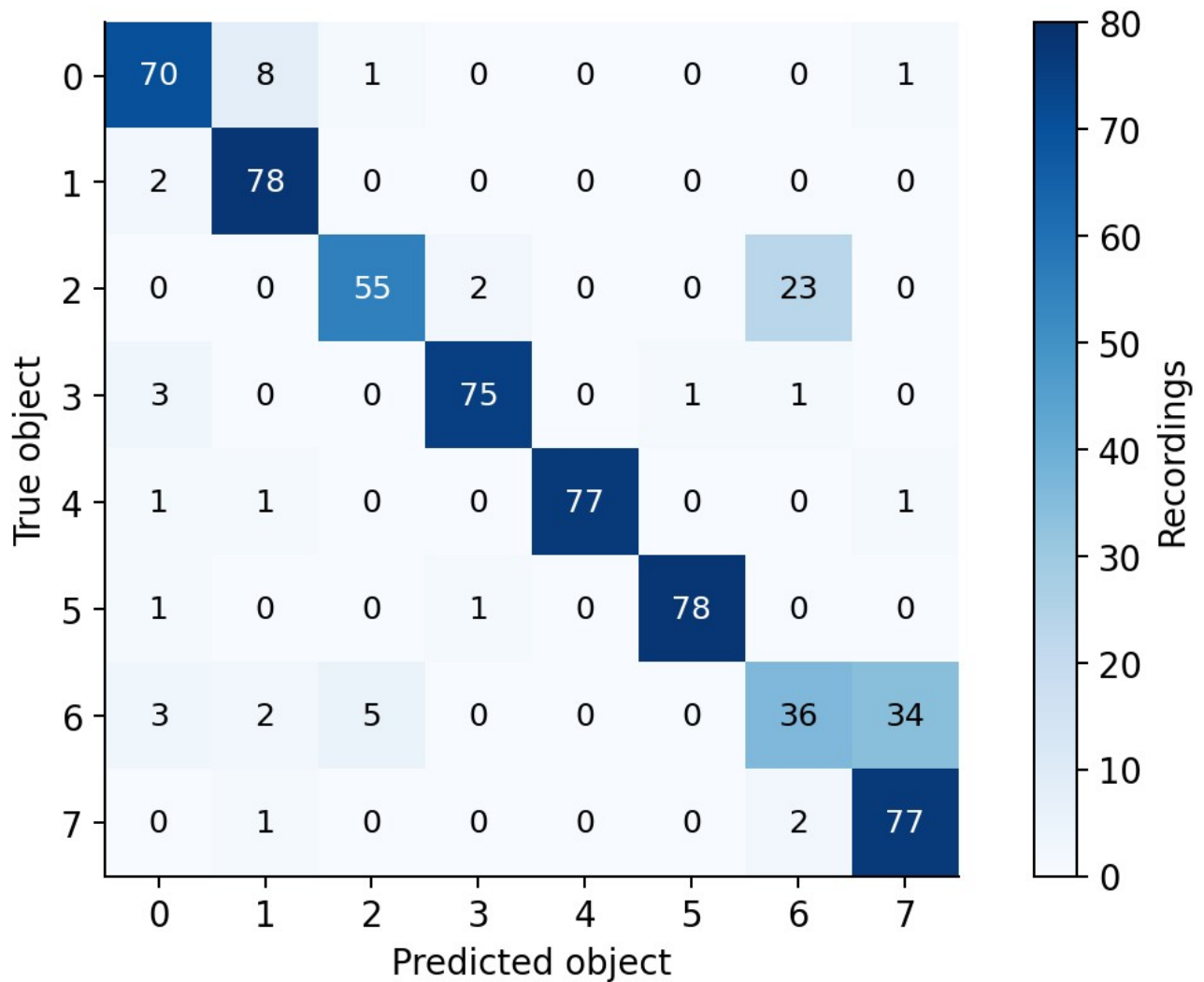


Figure 2. Final pooled-model confusion matrix. Rows are the true object and columns the predicted object. Diagonal cells are correct decisions; off-diagonal cells show where mistakes go.

A confusion matrix adds information that a single accuracy number hides. An object with low recall may be confused mainly with one alternative or spread across several alternatives. The numbers here are recordings, so each row sums to 80. The matrix is descriptive; it does not explain why a pair is confused.

The earlier 96.25% model was trained on repetitions from B1–B4 and tested on another repetition within those same blocks. The final 85.31% model was trained on B0 and tested across B1–B4. These differ in training data, evaluation task and movement preparation. Their difference is not a controlled estimate of the effect of changing object locations.

Likewise, subtracting 85.31% from the 99.06% B0 training score compares fit with transfer, not two independent test domains. It shows that fitting B0 is easier than this evaluation, but it is not a statistical significance test.

The useful interpretation is therefore specific: with the agreed preprocessing, one fixed training seed and the B0-to-B1–B4 split, the model gets 546 of 640 movements right. Object-wise and participant-wise results are needed to understand that average.

How much movement is needed

The first anticipation experiment keeps the trained model frozen and supplies increasingly long prefixes of each test movement. These are the first positions of the 100-position resampled trajectory, not a separately retrained model at every fraction.

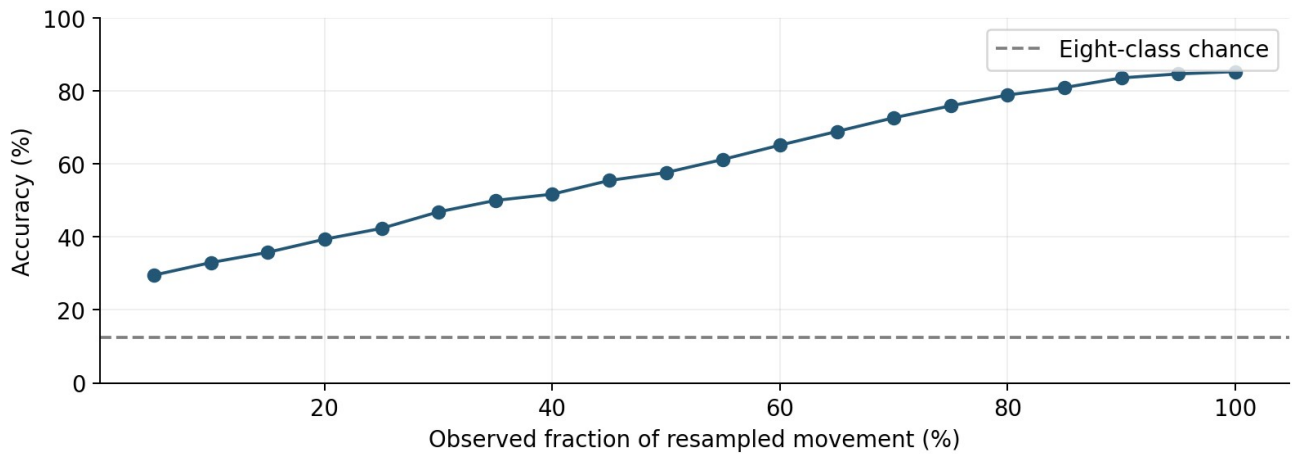


Figure 3. Accuracy over the 5%-spaced observation grid. All 640 test recordings are evaluated at every fraction. The dashed line is balanced eight-class chance, 12.5%.

Movement observed	5%	10%	25%	50%	75%	100%
Accuracy	29.5%	33.0%	42.3%	57.7%	75.9%	85.3%

Accuracy rises as more movement becomes available. This does not mean the early movement contains only as much information as this full-trained model can use: the earlier matched-prefix experiments show that training specifically on short inputs can change the result.

A second summary asks when a recording first becomes correct and stays correct at every later evaluated fraction. Among the 546 recordings correct at full length, the median first sustained fraction is 30% and the mean is about 37%. The remaining 94 recordings cannot meet this criterion because their final prediction is wrong.

This is a retrospective, label-based summary on a discrete grid. It does not mean that the model knows at 30% that it should stop, and it does not describe all trials. A fraction also has no single duration in milliseconds: movements have different lengths. That is why the next experiment returns to the original timestamps.

Anticipation in physical time

For each recording, the input is cut at specified times before the sensor-defined lift event. Only glove observations at or before that cutoff are used. Prefixes with fewer than two observations are excluded, so the number of eligible recordings is reported.

Two representations are compared: interpolate the available prefix back to 100 positions, or feed its raw, variable number of observations directly. The CTM accepts the latter without padding or masking. The model and B0 scaling remain frozen in both conditions.

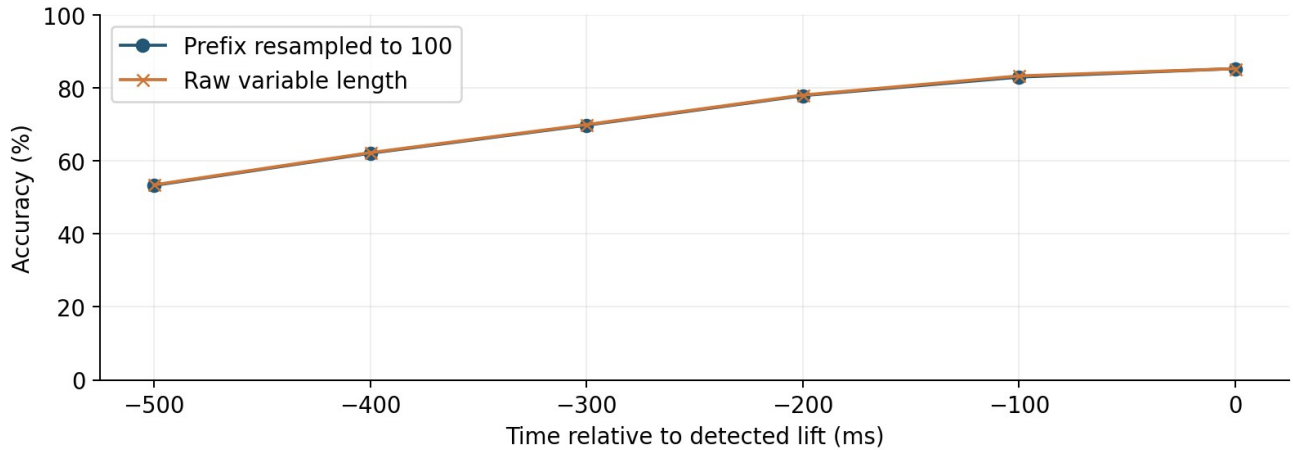


Figure 4. Pooled-model anticipation with and without resampling each prefix. The curves nearly overlap. Zero is the detected lift reference; negative values are earlier observations.

Lead before lift	Valid recordings	Mean raw frames	Resampled accuracy	Raw accuracy
500	608	31.1	53.29%	53.45%
400	615	36.9	62.11%	62.28%
300	626	42.6	69.81%	69.97%
200	629	49.0	77.90%	78.06%
100	629	55.0	82.99%	83.31%
0	640	60.9	85.31%	85.31%

The largest pooled accuracy difference between representations is only about 0.32 percentage points. The anticipation result therefore does not appear to depend materially on stretching short inputs to 100 positions in this run. At 500 ms, raw-input accuracy is 53.45%; at 100 ms it is 83.31%.

A fixed set of 608 recordings eligible at 500 ms gives raw accuracies of 53.45%, 62.17%, 69.74%, 77.96%, 83.22% and 85.03% across the six cutoffs. The same rising pattern remains when cohort composition is held constant. Actual observations fall slightly before nominal cutoffs because the glove is sampled roughly every 16 ms.

Some objects become recognizable earlier

The overall anticipation curve combines quite different object patterns. The object-level view below uses the original resampled-to-100 physical-time evaluation. Each cutoff includes the recordings eligible at that time, so early denominators can differ between objects.

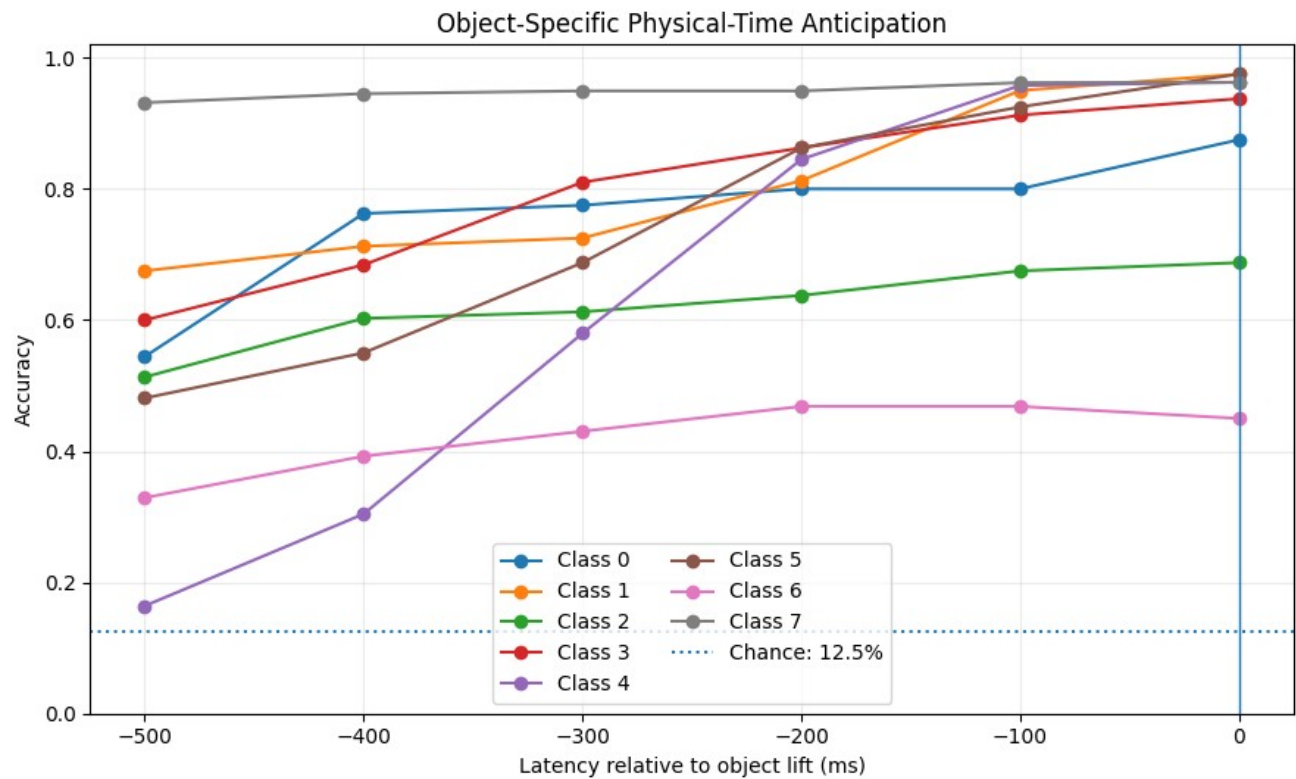


Figure 5. Object-specific anticipation from the saved notebook plot. Accuracy is shown from zero to one, so 0.8 means 80%. This figure uses resampled prefixes; the shorter-input check is reported separately.

Class 7 is already recognized at about 93% accuracy 500 ms before lift. Class 4 starts around 16%, then rises to 96% by lift. In contrast, class 6 stays below 50% across the entire interval. These curves show why “how early can the model recognize an object?” has no single answer for the whole dataset.

Lead before lift	500 ms	400 ms	300 ms	200 ms	100 ms	0 ms
Class 4 recall	16.4%	30.4%	58.0%	84.5%	95.8%	96.3%

A useful distinction is late emergence versus persistent difficulty. Class 4 becomes much easier with later movement, whereas class 6 remains hard even when the full crop is available. More observation helps the first pattern considerably; it does not resolve the second under this trained model.

These are recognition patterns rather than proof of when intention forms in a person. The model may use hand shape, position or other coordinate regularities. The present experiments do not isolate which of those features causes each curve. The object identifiers are kept numeric rather than assigning unverified object names.

One pooled model or one per person

Each participant-specific model is trained only on that person's B0 recordings, with its own B0-fitted scaler, then tested on the same person's B1–B4. The architecture, seed and 150-update budget are kept the same. With 63–64 training recordings, this takes 75 epochs rather than the pooled model's 15.

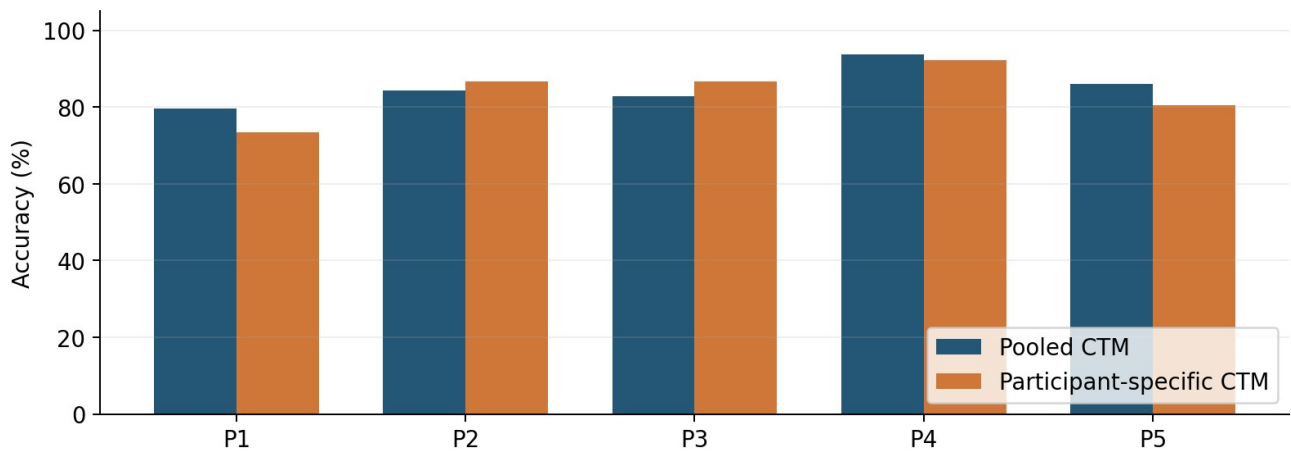


Figure 6. Full-movement transfer for each participant. Each pair of bars evaluates the same 128 recordings. These are individual participant results, not variability across random training seeds.

Participant	B0 records	Pooled correct	Personal correct	Personal accuracy
P1	64	102 / 128	94 / 128	73.44%
P2	63	108 / 128	111 / 128	86.72%
P3	63	106 / 128	111 / 128	86.72%
P4	64	120 / 128	118 / 128	92.19%
P5	64	110 / 128	103 / 128	80.47%

Personalization improves P2 and P3, but reduces performance for P1, P4 and P5. Across all test recordings it gives 537/640 correct, versus 546/640 for the pooled model. All personal models fit their B0 training recordings perfectly, which again does not guarantee transfer.

A plausible trade-off is specialization versus training diversity: a personal model focuses on one person but loses examples from the others. This is an interpretation, not a tested causal explanation. Equal update counts also mean different numbers of passes through the available data; they do not make every aspect of learning identical.

Both protocols contain the tested participant in training. Neither is an unseen-person test. Their difference is modest and comes from one seed, so it does not establish that pooled models are generally superior.

Personal models before object lift

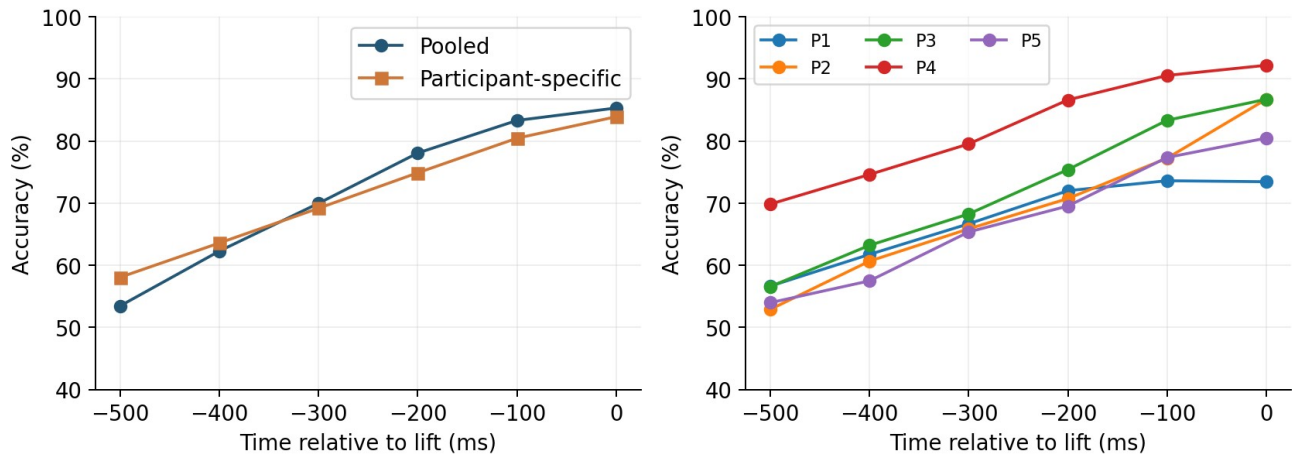


Figure 7. Raw-input physical-time anticipation. Left compares recording-weighted totals; right shows the five personal models separately. Eligible counts match the pooled evaluation at each cutoff.

Lead before lift	Pooled accuracy	Personal models	Difference in pp
500	53.45%	58.06%	+4.61
400	62.28%	63.58%	+1.30
300	69.97%	69.17%	-0.80
200	78.06%	74.88%	-3.18
100	83.31%	80.45%	-2.86
0	85.31%	83.91%	-1.41

The personal models are somewhat better in aggregate at the earliest cutoffs, but the pooled model is better from 300 ms onward. A full-trajectory ranking therefore does not tell the whole anticipation story. The participant curves also show why the average should not stand alone.

Within the personal models, raw-input and resampled accuracy usually differ by less than one percentage point. The largest participant/cutoff difference is about 1.63 points. Prediction disagreement is below 1% when pooled across participants at each lead time, but individual participant/cutoff cells reach about 1.77%. Similar overall accuracy does not mean every prediction is identical.

These results are comparisons at predefined cutoffs relative to a known future lift event. They demonstrate recognition from incomplete movement, not an online commitment rule or a measured reaction time. The model is rerun on each available prefix; this report does not claim a continuously streaming deployment.

Earlier experiments and the overall conclusion

The earlier repetition-split studies used B1–B4 repetitions 0–1 for training, repetition 2 for validation and repetition 3 for testing. They are retained as supporting observations, separate from the final transfer protocol.

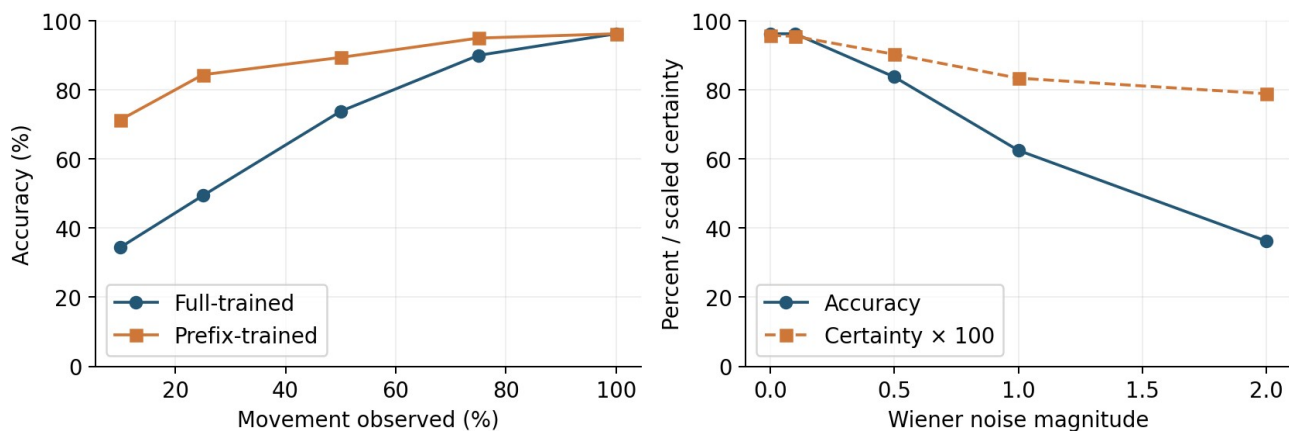


Figure 8. Earlier held-out-repetition results, with 160 test recordings. Left compares full-trained and prefix-trained models. Right shows accuracy and native certainty under correlated input noise. These are not final B0-transfer results.

Earlier experiment	Observation
Full trajectory	154 / 160 correct 96.25%
Matched 10% prefix	71.3% versus 34.4% from the full-trained model
Equal quarter-length windows	Beginning 84.4% middle 93.8% end 96.9%
Internal ticks on a 10% prefix	About 70–73%; no consistent gain from 5 to 25 ticks
Unseen participant pilot	Mean 75.94%; individual results 49.22–98.44%

The strongest completed finding is that the B0-trained CTM recognizes targets across later blocks and gains accuracy as movement unfolds. Short raw inputs preserve this pattern. Transfer varies by object and person, and personalization is not a uniform improvement. Small differences remain descriptive because repeated training seeds and confirmatory model comparisons were not performed.

The saved checkpoints, scalars, recording identities and per-tick outputs make the next analysis possible: inspect how the already trained model changes internally. That asks a different question from recognition accuracy. It does not require retraining the classifier or assuming that its ticks already implement evidence accumulation.

A short way to explain the work: “I trained CTM on the first block and tested the later blocks. It recognized most movements, with errors concentrated in a few objects. Recognition improved before lift, and that result survived using shorter raw inputs. Personal models helped some participants but not all. I then preserved these models to study their internal decision dynamics.”

Evidence: 03_rtg_experiments.ipynb, 178-cell snapshot reviewed on 30 September 2026; rtg_experiment_runner.py and data_loader.py for representation definitions; pooled run 20260927T130855Z_final_all_b0_to_b1_b4_seed0; follow-up run 20260928T123450Z_rtg_supervisor_followup and five participant-specific fixed150 runs. Tables were checked against notebook outputs and saved predictions. Figures are redrawn from those results or retained from the notebook. Code and routine checks are omitted.